Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	25	**1**	
1(a)(ii)	$\frac{5}{8}$ cao	**2**	**M1** for $\frac{225}{360}$ oe
1(a)(iii)	1:2:5	**2**	**B1** for 45 : 90 : 225 oe or better
1(a)(iv)	Correct explanation eg if chemistry has an odd number then biology would not be a whole number of students	**1**	
1(a)(v)	65	**2**	**M1** for $\frac{90}{26}=\frac{225}{x}$ oe or better or **M1FT** *their* **(a)(iii)** for $\frac{their5}{their2}\times 26$
1(b)(i)	28	**1**	
1(b)(ii)	14	**1**	
1(b)(iii)	$\frac{9}{28}$ oe	**1**	**FT** $\frac{9}{their\textbf{(b)(i)}}$
1(b)(iv)	6, 8, 10, 4	**1**	
2(a)	8.7[0]	**3**	**M2** for $3.5\times1.2+2\times2.25$ oe or **B2** for 4.2[0] and 4.5[0] or **M1** for 3.5×1.2 oe or 2×2.25 oe
2(b)	1.68 cao	**1**	
2(c)	2.07 cao	**2**	**M1** for $\left(1+\frac{15}{100}\right)\times1.8$ oe or **B1** for 0.27
2(d)(i)	$5+10\times1.5$ [=20]	**1**	

Question	Answer	Marks	Partial Marks
2(d)(ii)	Ruled line from (0, 5) to (10, 20)	**2**	**B1** for one correct point plotted
2(e)	12 1.4[0]	**3**	**M2** for $20-12\times1.55$ oe or **B2** for 18.6 or **M1** for $\frac{20}{1.55}$ soi by 12.9… or 12 in the answer space
2(f)	300	**2**	**B1** for 300*k* as final answer or **M1** for [12 =] 2 × 2 × 3 **and** [50 =] 2 × 5 × 5 or 2 correct factor trees/lists/tables or a list of multiples of both 12 and 50 with at least 3 of each or 2 × 2 × 3 × 5 × 5 or 2,2,3,5,5 or $50\times\frac{12}{50-(12\times4)}$ oe
2(g)	*A* With correct comparisons made of the 3 bottles with suitable accuracy shown	**3**	**M2** for 3 correct comparable values, or for a correct method to compare 3 bottles shown but not evaluated to enough accuracy or **M1** for 2 correct comparable values or for a correct method to compare 3 bottles but not evaluated
3(a)	9.5	**1**	
3(b)	38	**1**	
3(c)	29	**1**	
3(d)(i)	48	**1**	
3(d)(ii)	Correct dimensions	**1**	
3(e)	352π cao nfww	**3**	**M2** for $\pi20^2-3\times\pi4^2$ oe or **M1** for $\pi20^2$ or $\pi4^2$ oe
3(f)(i)	140	**1**	